“Well, let’s not get too full of ourselves just yet,” her father muttered, seemingly returning to his skeptical stance. He lifted the toothpick from his glass and bit off an olive. “Let me try to understand what the theory actually says. Is it right to say that the number of worlds being produced at every moment is literally infinitely big?”

“Well,” replied Alice, somewhat tentatively, “I’m afraid an honest answer to that question is going to require a bit more philosophical hair-splitting.”

“Why am I not surprised?”

“We can go back to entropy as an analogy. When Boltzmann came up with his entropy formula, he counted the number of microscopic arrangements of a system that looked macroscopically the same. From there, he was able to argue that entropy should naturally increase.”

“Sure,” said her father. “But that is real, honest physics, something we can test experimentally. Not sure what it has to do with your Many-Worlds flights of fancy.”

“We say that now. But you have to imagine what people were thinking back at the time.” Alice was settling comfortably into professor mode, her Bordeaux momentarily forgotten. “Boltzmann was right, but a number of objections were raised to his idea. One was that he was turning entropy from an objective feature of a physical system into a subjective one, which depended on some notion of ‘looks the same.’ Another was that he demoted the second law from an absolute statement to a mere tendency—it wasn’t that entropy necessarily increased, it was just very likely to do so. Particles jiggle around randomly, and it’s extremely probable that they will evolve toward a higher-entropy state, but it’s not a lawlike certainty. With the wisdom of accumulated years, we can see that the subjective nature of Boltzmann’s definition does not stop it from being a useful one, and the fact that the second law is a really good approximation rather than an absolute unbreakable law is more than good enough for whatever purposes we may have.”

“I get that,” answered her father. “Entropy is an objectively real thing, but we can define and measure it only after making a few decisions. But that never really bothered me—it’s useful! I’m not sure that extra worlds really are.”

“We’ll get there, but first let me elaborate on this analogy. Like entropy, the notion of a ‘world’ in Everettian quantum mechanics is a higher-level